

EE5203 L05 Pre-Lab Quiz

HAL GPIO and the B1 button - Basri Erdogan

Preparation time: 5-8 minutes **Attempts:** Two; highest score recorded **Bring to the lab:** Your answer to Question 5

1. What does GPIO_MODE_OUTPUT_PP make HAL_GPIO_Init write for PA5?
 - a. MODER[11:10] = 01 and OTYPER[5] = 0
 - b. MODER[11:10] = 10 and OTYPER[5] = 1
 - c. BSRR = (1U << 5)
 - d. PUPDR[11:10] = 01
2. Why does HAL_GPIO_Init(GPIOA, &config) receive the **address** of config?
 - a. Structures cannot be used as arguments in C.
 - b. The function reads the members through the pointer instead of receiving a copy.
 - c. & marks the argument as unchangeable.
 - d. HAL requires all arguments to be addresses.
3. On the Nucleo board, B1 is pressed. What does HAL_GPIO_ReadPin(GPIOC, GPIO_PIN_13) return, and why?
 - a. GPIO_PIN_SET — the button connects PC13 to 3.3 V.
 - b. GPIO_PIN_SET — the internal pull-up holds the pin high.
 - c. GPIO_PIN_RESET — the button connects PC13 to ground; the external pull-up only holds it high while released.
 - d. GPIO_PIN_RESET — inputs always read low.
4. Changing config.Speed from GPIO_SPEED_FREQ_LOW to GPIO_SPEED_FREQ_HIGH will:
 - a. make the LED blink faster.
 - b. change the pin's electrical transition strength only — the blink timing still comes from HAL_Delay.
 - c. shorten HAL_Delay(500) to HAL_Delay(50).
 - d. change MODER to 10.
5. Complete this entry ticket before class:

Prediction	Your answer
GPIOC->MODER bits for PC13 input, and their value	
GPIOC->PUPDR bits 27:26 with GPIO_NOPULL, and why no pull is correct here	
The falling-edge condition (using prev and cur) that counts one press	
First SFR check if the LED never blinks though the build is clean	